

Suman Jash

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SUMMARY

Computer Science Engineering student who designs, builds, and deploys full-stack, backend, and machine learning systems using Python, FastAPI, Flask, SQL, Docker, and GitHub Actions. Strong foundation in system design, database schema design, authentication and security, REST APIs, and machine learning model evaluation, with experience shipping projects end-to-end from design through CI/CD and deployment. Codeforces Pupil (1330), CodeChef 3-Star (1628), 600+ problems solved.

EDUCATION

Narula Institute of Technology

B.Tech in Computer Science and Engineering, CGPA: 8.96/10

2023 – 2027

Kolkata, West Bengal

Arambagh Vivekananda Academy

CBSE Class XII – 85.8% | CBSE Class X – 94.4%

2021 – 2023

Arambagh, Hooghly

PROJECTS

- **ShetBhav – Market Intelligence Platform for Farmers** | *Python, FastAPI, Next.js, TypeScript, PostgreSQL, XGBoost*
[GitHub](#) | [Live Demo](#)
 - Designed and built a full-stack platform with 104 REST endpoints, a 45-table PostgreSQL schema, and 24 Next.js/TypeScript routes, supporting 4 user roles with server-side RBAC.
 - Built a multi-factor recommendation engine (8 weighted factors) and an XGBoost price-forecasting pipeline integrated with live market data from AGMARKNET (data.gov.in).
 - Designed a full marketplace transaction flow – listings, offers, counter-negotiation, and order tracking – with JWT authentication and bcrypt password hashing.
 - Wrote 246 backend tests and 15 end-to-end tests in a GitHub Actions CI/CD pipeline; deployed on Vercel and Render with PostgreSQL.
- **CourierAI – ML-Powered Delivery Time Prediction Platform** | *Python, Flask, scikit-learn, SQLite, Docker*
[GitHub](#) | [Live Demo](#)
 - Tuned a HistGradientBoostingRegressor on 49,639 real courier records via randomized search over 150 configurations, reaching 0.5361-day MAE and 0.7466 R^2 under 5-fold cross-validation.
 - Built a benchmarking harness comparing 5 base models (Random Forest, XGBoost, CatBoost, SVR, MLP) against voting and stacking ensembles for regression and classification.
 - Designed a secured REST API with encrypted per-user API keys, tiered rate limiting, and plan-based quotas; containerized with Docker and deployed via an automated GitHub Actions CI/CD pipeline.
- **AeroBook – Airline Reservation & Flight Operations Platform** | *PHP 8, MySQL, Bootstrap 5, JavaScript, Docker*
[GitHub](#) | [Live Demo](#)
 - Designed a normalized 24-table MySQL schema (FK constraints, 13 indexes) powering 40+ pages across passenger and admin operations.
 - Built a Smart Fare Engine that discovers connecting itineraries in 3 DB queries per search, ranking each by price, duration, layover, and stop count.
 - Hardened the app with prepared statements, CSRF protection, and bcrypt hashing; wrote a dependency-free smoke-test suite and shipped a containerized Docker deployment.
 - Integrated live airport and airline data sync from the AviationStack API into an admin analytics suite with CSV exports and a searchable audit trail.

TECHNICAL SKILLS

Languages: C++, Python, Java, PHP, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: FastAPI, Flask, Next.js, scikit-learn, Pandas, NumPy, XGBoost, CatBoost

Databases: PostgreSQL, MySQL, SQLite

Tools: Git, GitHub Actions, Docker, Docker Compose, pytest

Deployment: Render, Vercel, Gunicorn

Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, REST APIs, Authentication, Web Security, CI/CD, Machine Learning

ACHIEVEMENTS

- Secured 2nd rank in *Code It*, a college-level duo-format coding competition with 40 teams competing in the final round.
- Led a team in the internal Smart India Hackathon round to build a College ERP System concept; owned system design, QA/testing, and project presentation after selection for the college-level demonstration round.